

S.No. : 551

BCS 3604

No. of Printed Pages : 05

Following Paper ID and Roll No. to be filled in your Answer Book.

PAPER ID : 33228

Roll
No.

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B. Tech. Examination, 2024-25

(Even Semester)

COMPILER DESIGN

Time : Three Hours]

[Maximum Marks : 60

Note :- Attempt all questions.

SECTION - A

1. Attempt all parts of the following : $8 \times 1 = 8$

- What are the operation required on a symbol table?
- Why loop optimization is so important than other code optimization?
- Differentiate compile and interpreter.
- Differentiate parse tree and annotated parse tree.

[P. T. O.

- (e) What is directed acyclic graph?
- (f) Specify the two function that the parsing table of LR parser can have.
- (g) Draw syntax tree for expression :
 $(a + b) * (a + b + c)$
- (h) What are the goal of error handler in parser?

SECTION - B

2. Attempt any two parts of the following : $2 \times 6 = 12$

- (a) What are the phases of the compiler? Explain each phase in detail.
- (b) Consider the grammar :

$S \rightarrow icts$
 $S \rightarrow ictses$
 $S \rightarrow a$
 $S \rightarrow b$

Construct a leftmost and rightmost derivation for the string : ibtibtaea

- (c) Generate three address code for the following statement assuming three registers are available:

$$x = \overbrace{a}^{t_1} / \overbrace{(b+c)}^{t_2} - d * \overbrace{(e+f)}^{t_4}$$

- (d) What do you understand by boot strapping? Explain with help of example.

SECTION - C

Note :- Attempt all questions. Attempt any two parts from each questions. $8 \times 5 = 40$

- 3. (a) Explain about basic parsing techniques. What is top down parsing? Explain in detail.
- (b) Discuss the role of syntax directed translation scheme. Describe the various code optimization technique in detail.
- (c) Explain activation tree in runtime environment.
- 4. (a) Consider the grammar with the following translation rules & E as the start symbol :

$E \rightarrow E1 + T \{ \text{print}(' + '); \}$

$E \rightarrow T$

$T \rightarrow T1 * F \{ \text{print}(' * '); \}$

$T \rightarrow F$

$F \rightarrow \text{num print}(' \text{num.val} '); \}$

Construct parse tree for the string $2 + 3 * 4$ find what will be printed.

[P.T.O.]

- (b) Construct the LL (1) parsing table for the following grammar?

$$E \rightarrow E+T/T$$

$$T \rightarrow T * F$$

$$F \rightarrow (E)/id$$

What happens if multiple entries occurring in your parsing table? Justify your answer.

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- (c) Differentiate between the top down parsing and bottom up parsing.

5. (a) Construct SLR parsing table for the following grammar :

$$E \rightarrow T + E / T$$

$$T \rightarrow id \text{ } \&\text{ } \&$$

- (b) Show that the following grammar is LL (1) but not SLR (1) :

$$S \rightarrow AaAb / BbBa$$

$$A \rightarrow E \text{ } | \text{ } \epsilon$$

$$B \rightarrow E \text{ } \quad B \rightarrow E / \epsilon$$

- (c) List the advantages and disadvantages of operator precedence parsing.

Construct the operator precedence parser :

$$E \rightarrow E+E$$

$$E \rightarrow E * E$$

$$E \rightarrow id$$

6. (a) Explain DAG representation of basic blocks with an example. Discuss the strategies for loop optimization and dead code elimination.
- (b) Define peephole optimization. List the characteristics of peephole optimization. Explain the main issues of code generation in detail.
- (c) Explain various run time storage allocation strategies. Give the general structure of activation record. Give the purpose of each component.
